

COMMONWEALTH OF AUSTRALIA

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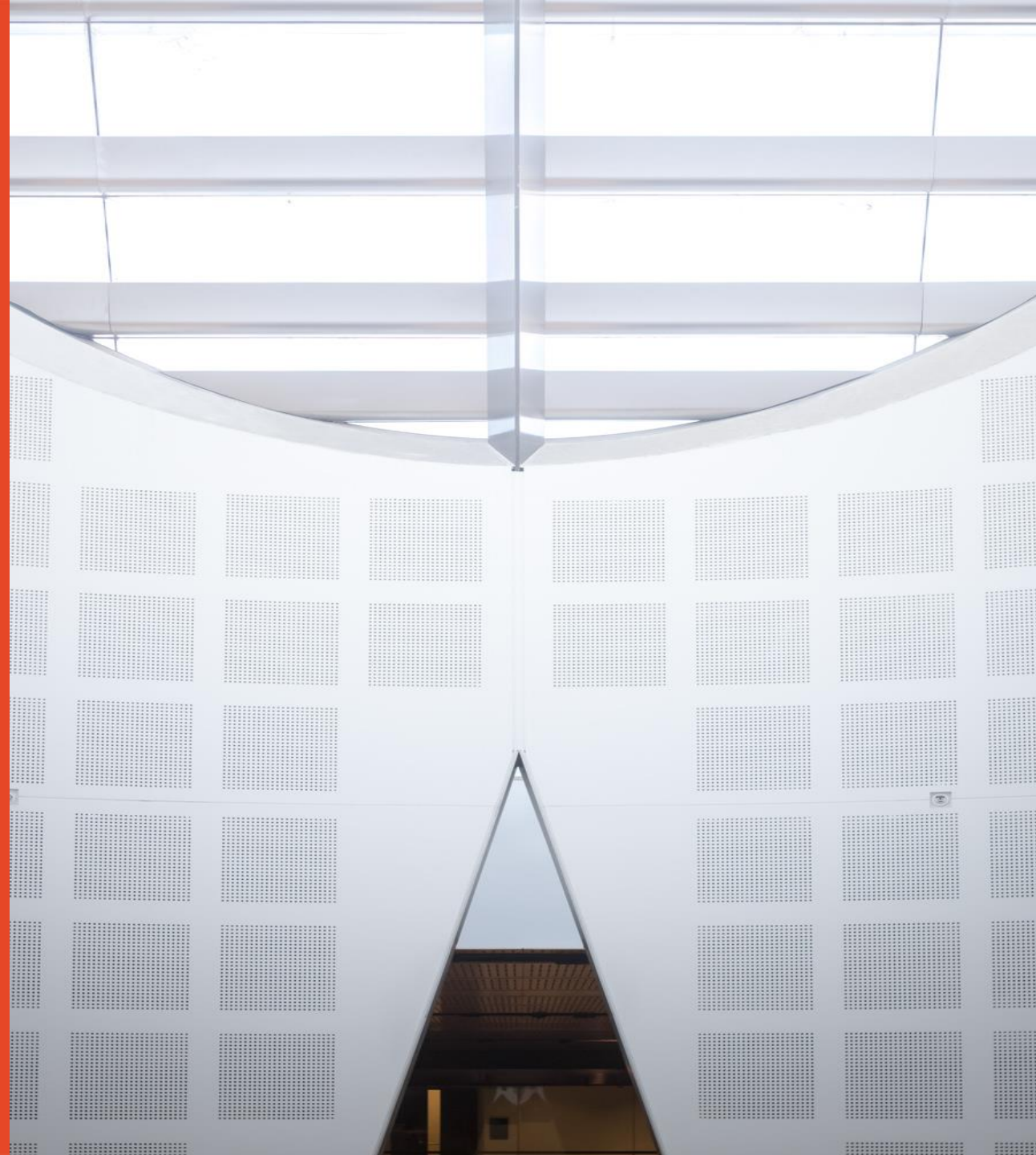
COMPx270: Randomised and Advanced Algorithms

Lecture 3: Balls in Bins

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THE UNIVERSITY OF
SYDNEY



A question 🎂

There are quite a few people in the classroom right now. What are the odds two of you (at least) have the same birthday?

A question 🎂

Theorem. (The 🎂 paradox) If you gather 23 people in a room, then with probability at least 50% a pair will sharing their birthday.

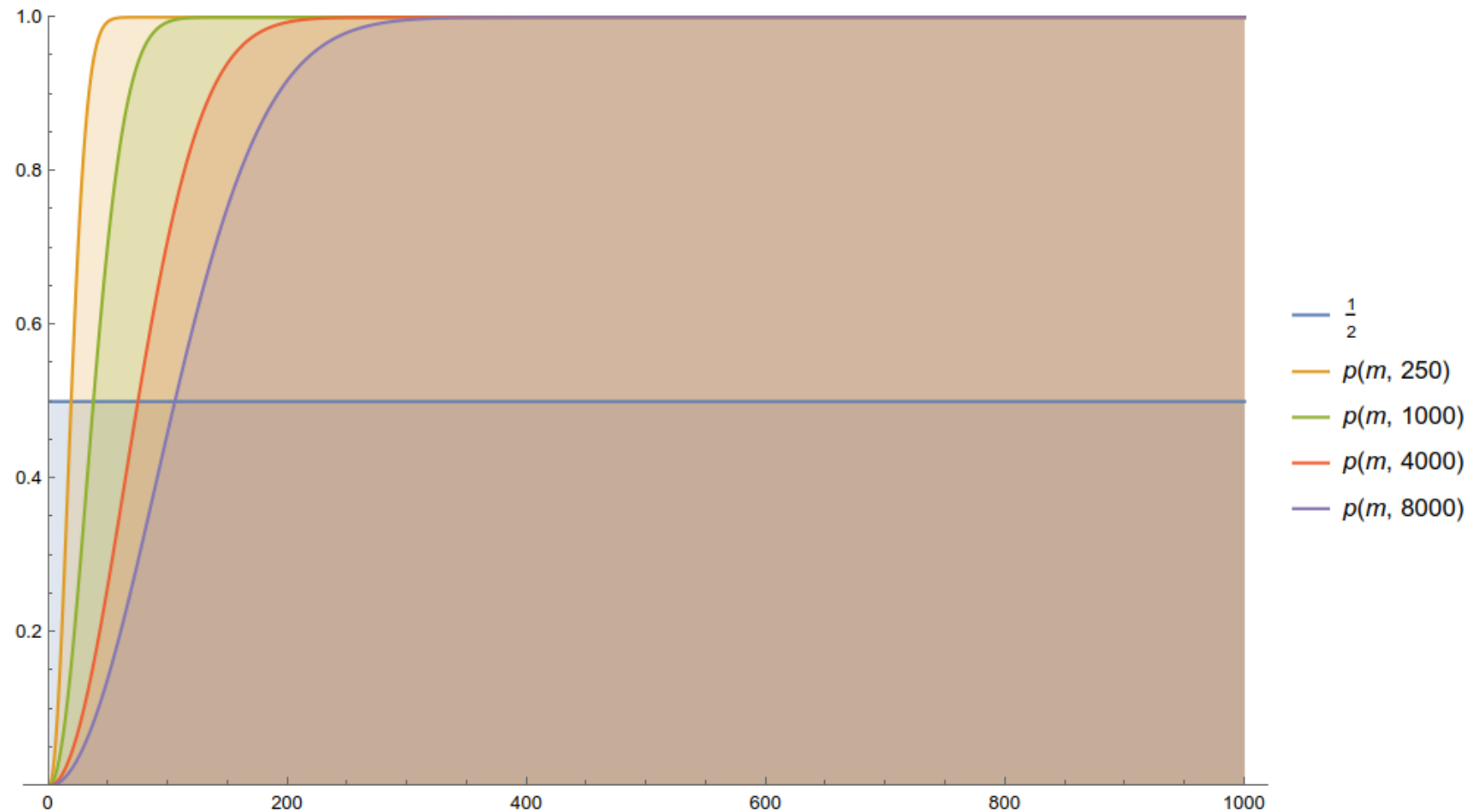
An answer? 🎂

Theorem. If you gather m people and give each a number uniform between 1 and n , then the probability $p(m, n)$ that at least two have the same number is...

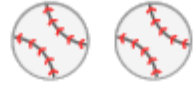
$$p_{m,n} = 1 - \frac{n!}{n^m (n - m)!} = 1 - \frac{m!}{n^m} \binom{n}{m}$$

Proof.

```
DiscretePlot[{1/2, p[m, 250], p[m, 1000], p[m, 4000], p[m, 8000]}, {m, 1, 1000}, PlotRange -> {0, 1}, PlotLegends -> "Expressions"]
```



Let's start simple: $m=2$



Now, for large values of "2"...

C: number of collisions when throwing m  into n . What is

$$c(m, n) = E[C]?$$

Number of collisions

... and what is $\text{Var}[C]$?



Number of collisions

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Number of collisions

$$\text{Var}[C] = \binom{m}{2} \frac{1}{n} \left(1 - \frac{1}{n} \right)$$

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Now, we can use Chebyshev:

$$\Pr[X = 0] \leq 1/2$$

for $m = \Omega(\sqrt{n})$.

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for $m = \Omega(\sqrt{n})$. Is it tight?

Number of collisions

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Now, we can use Chebyshev:

$$\Pr[X = 0] \leq 1/2$$

for $m = \Omega(\sqrt{n})$.

By Markov, we also have

$$\Pr[X \neq 0] = \Pr[X \geq 1] \leq E[X] \leq 1/2$$

for $m = O(\sqrt{n})$.

Applications?



 Articles

About 8,580 results (0.13 sec)

Bounding the variance: is it always that bad?

Two tricks (and even 3).

Coverage (Coupon collector)



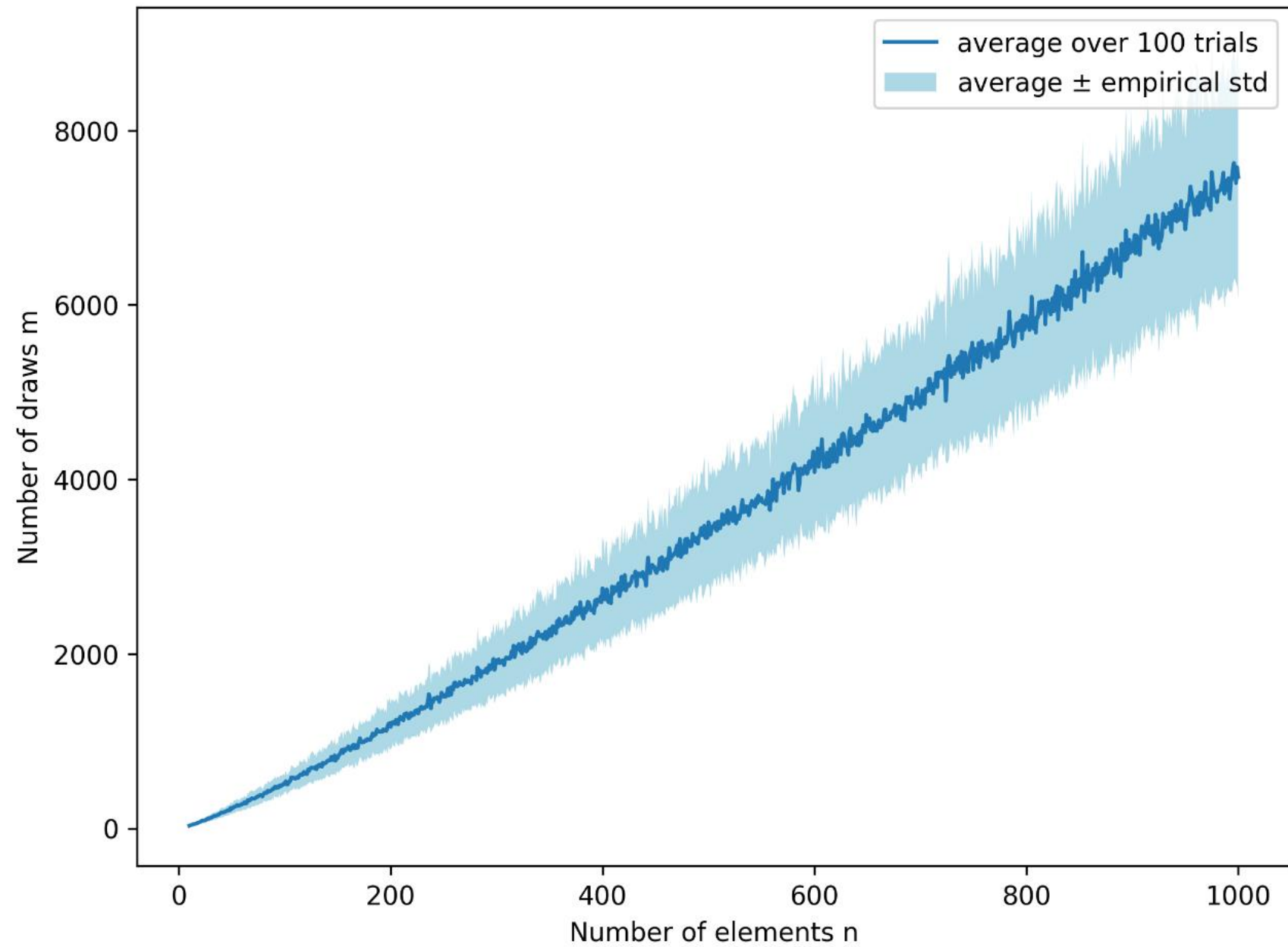
"What is the expected number of balls $M(n)$ we need to throw before each of the n bins contains **at least one** ball?"

Coverage (Coupon collector)



"What is the expected number of balls $M(n)$ we need to throw before each of the n bins contains **at least one** ball?"

- $\Theta(n)$?
- $\Theta(n \log n)$?
- $\Theta(n^2)$?
- Something else?



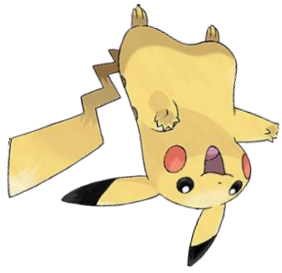
Coverage (Coupon collector)



"What is the expected number of balls $M(n)$ we need to throw before each of the n bins contains **at least one** ball?"

Theorem. In expectation, $M(n) = \Theta(n \log n)$ balls. (Even more precisely: $n \ln n + O(n)$.)

Proof.



What about the variance?



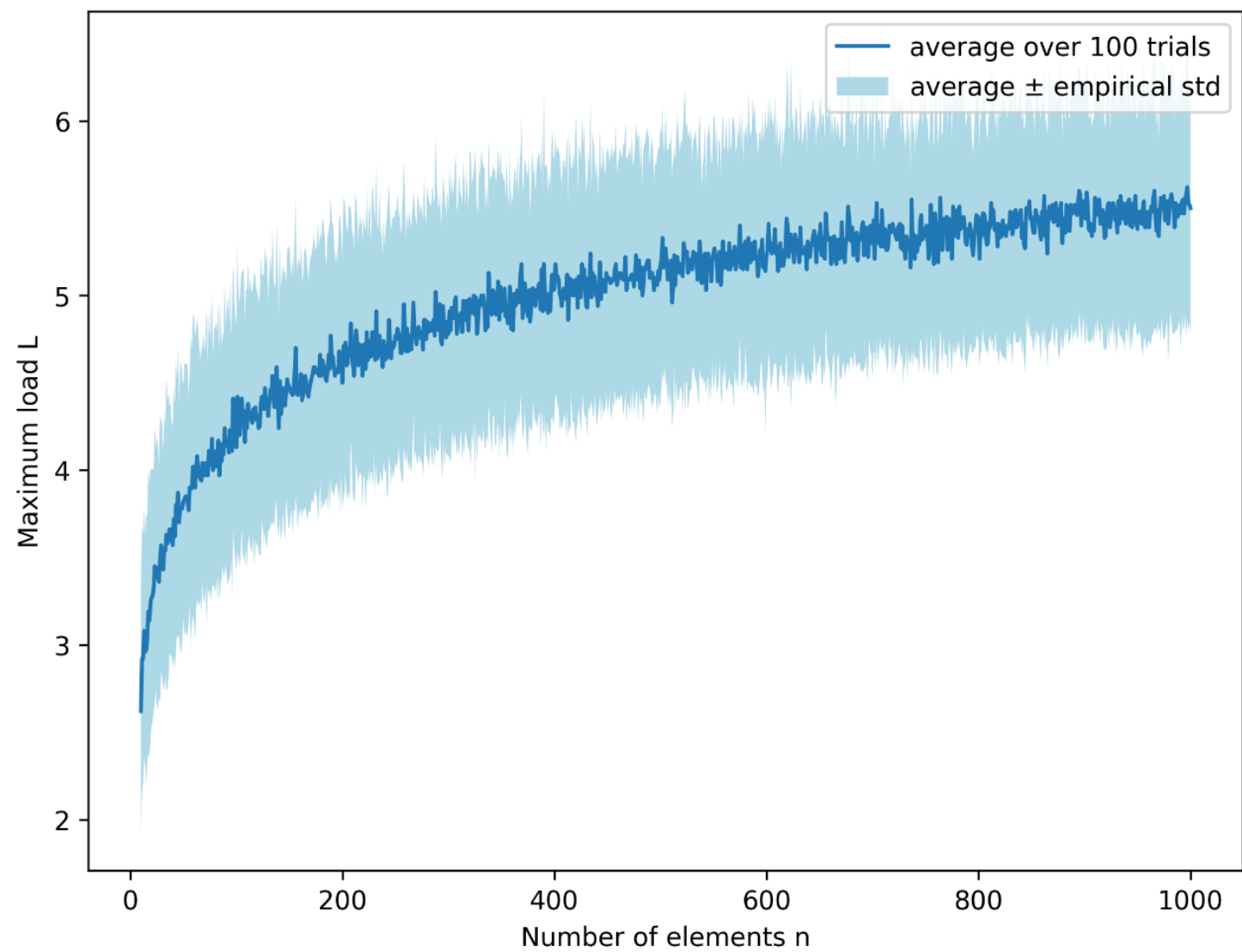
Load balancing

"What is the expected number of balls $L(n)$ the fullest of the n bins contains after throwing n balls?"

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- $\Theta(1)$?
- $\Theta(\log n)$?
- $\Theta(\sqrt{n})$?
- Something else?



Load balancing

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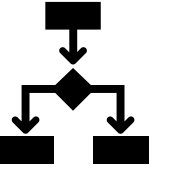
Theorem. The expected maximum load is $L(n) = \Theta(\log n / \log \log n)$.

Proof.

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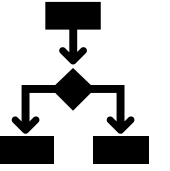
$$\binom{n}{k} \leq \left(\frac{e \cdot n}{k}\right)^k$$

Load balancing (a twist)



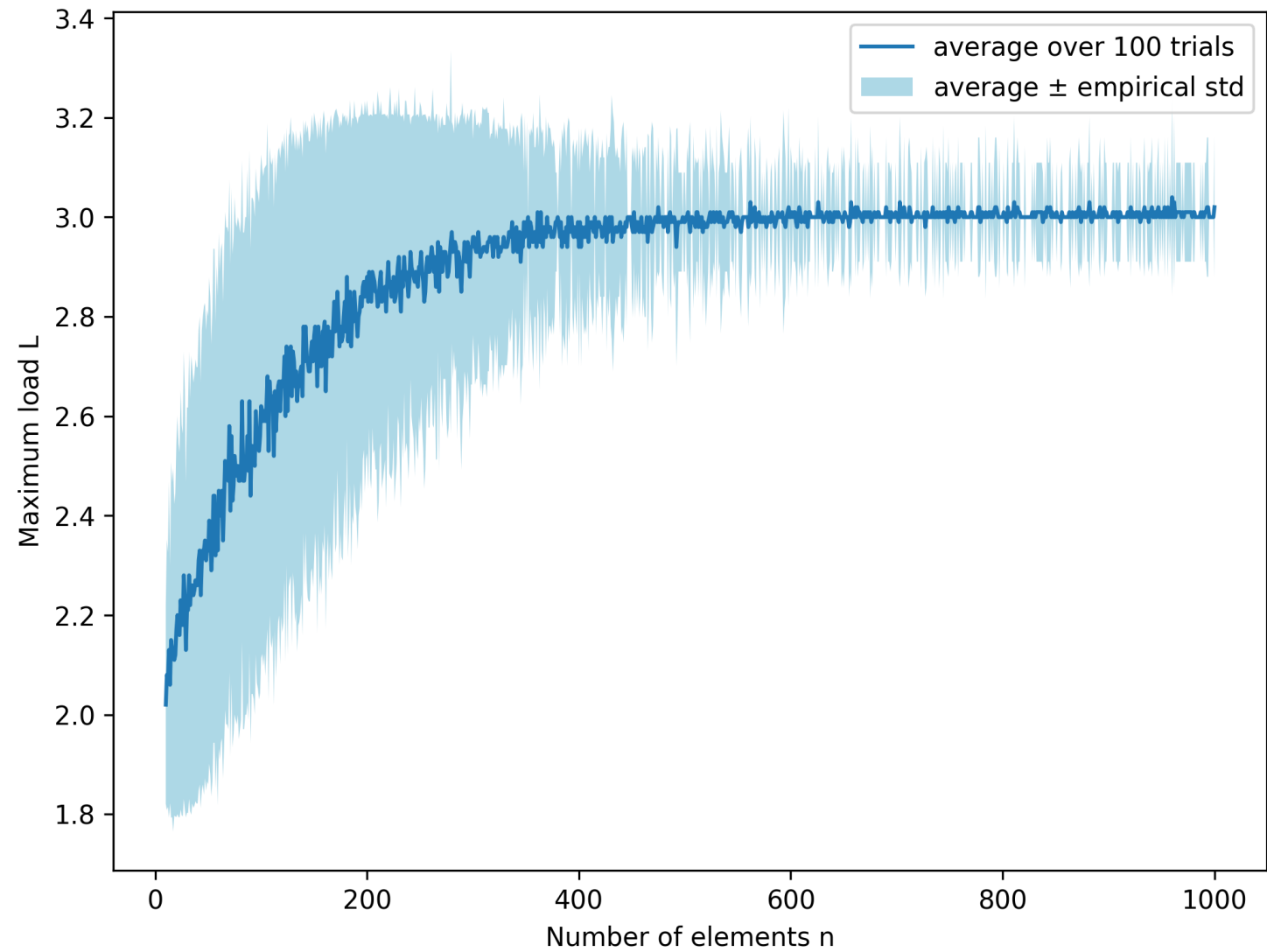
"Now, every time you throw a ball, it selects **two** bins at random, and goes to the least full of the two. What is the maximum expected load?"

Load balancing (a twist)

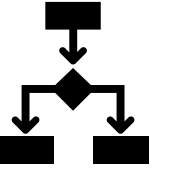


"Now, every time you throw a ball, it selects **two** bins at random, and goes to the least full of the two. What is the maximum expected load?"

- $\Theta(1)$?
- $\Theta(\log \log \mathbf{n})$?
- $\Theta(\sqrt{\log \mathbf{n}})$?
- Something else?

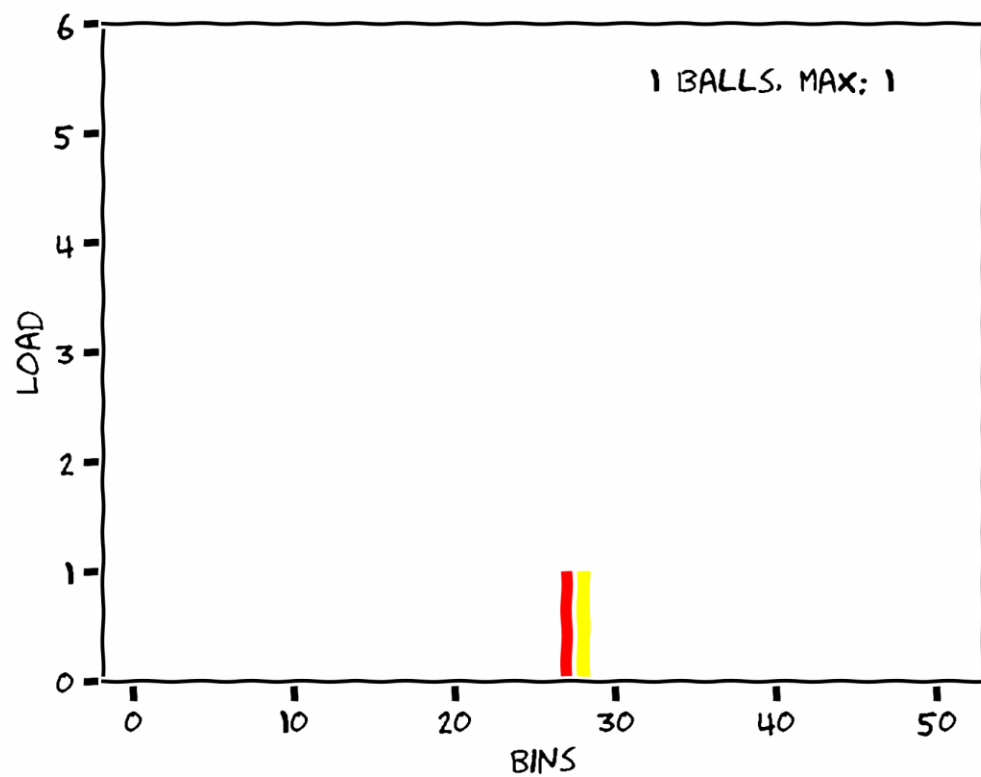


Load balancing (a twist)

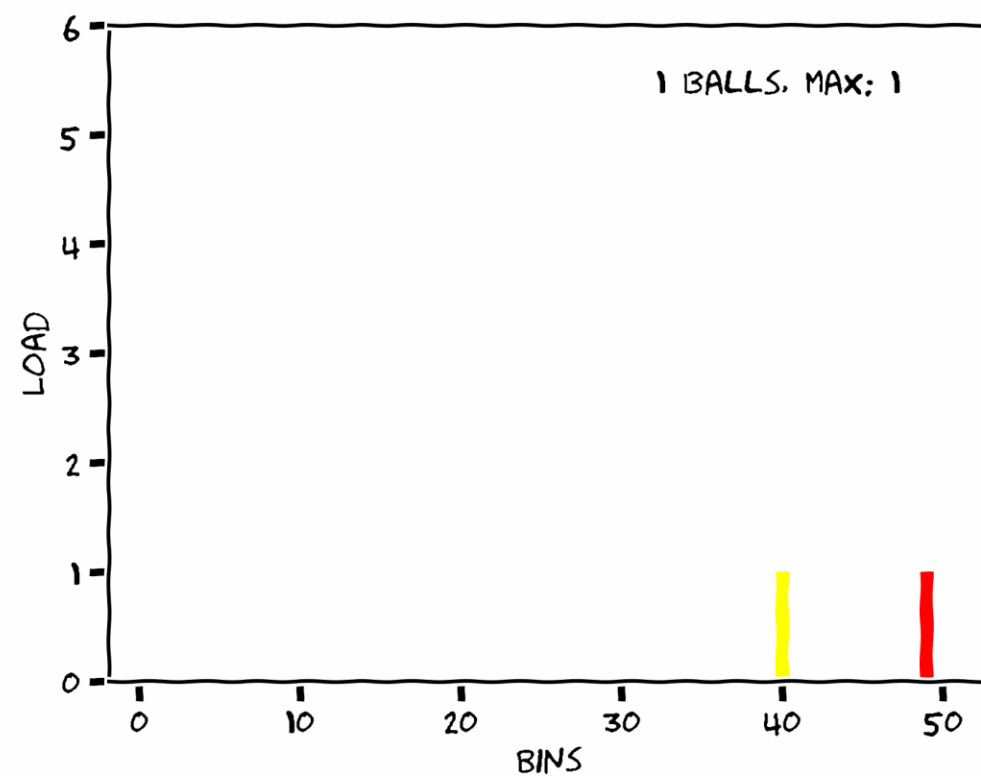


"Now, every time you throw a ball, it selects **two** bins at random, and goes to the least full of the two. What is the maximum expected load?"

Theorem. The expected maximum load now $\Theta(\log \log n)$.



"Standard" Balls in Bins



Power of Two Choices

Conclusion

